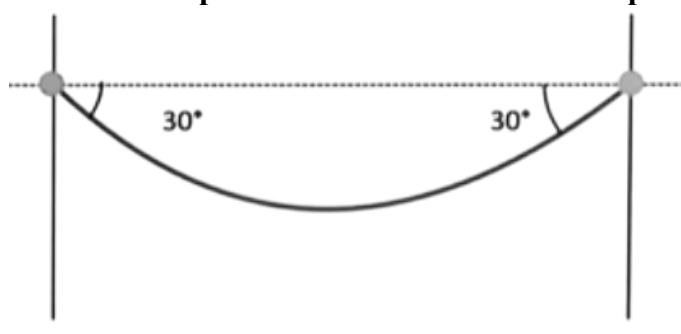


JEE-Main-24-01-2026 (Memory Based)

[EVENING SHIFT]

Physics

Question: A uniform rope is supported by two level pin support as shown in the figure. Mass of the rope is m . Find the tension at midpoint.

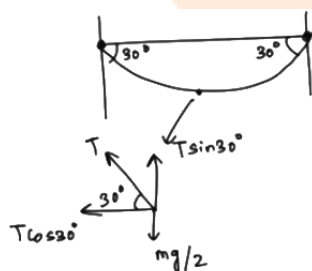


Options:

- (a) mg
- (b) $\frac{mg\sqrt{3}}{2}$
- (c) $\frac{4}{mg}$
- (d) $\frac{2}{mg}$

Answer: (b)

Solution:



$$T \sin 30^\circ = \frac{mg}{2} \quad \text{--- (i)}$$

$$T' = T \cos 30^\circ \quad \text{--- (ii)}$$

$$\frac{T'}{\frac{mg}{2}} = \cot 30^\circ$$

$$T' = \frac{mg}{2} \times \sqrt{3} \Rightarrow \left(\frac{mg\sqrt{3}}{2} \right)$$

Question: Find force on charge $q = 1 \mu\text{C}$ as due to uniformly charged rod as shown in the figure.



Options:

- (a) 7.5 N
- (b) 20 N
- (c) 2 N
- (d) 18 N

Answer: (a)

Solution:

Handwritten solution showing the calculation of the electrostatic force. It starts with the linear charge density $\lambda = \frac{2 \mu\text{C}}{0.1} = 20 \mu\text{C/m}$. Then, it shows a diagram of a small element dx at a distance x from the point charge $q = 1 \mu\text{C}$. The force on this element is $dF = k \frac{\lambda dx \cdot q}{x^2}$. The total force is calculated by integrating from $x = 0.02 \text{ m}$ to $x = 0.12 \text{ m}$, resulting in $F = 7.5 \text{ N}$.

Question: In a vernier callipers; vernier scale has 50 divisions & they coincide with 48 main scale divisions. Then what is the least count of vernier callipers?

Options:

- (a) 0.4 mm
- (b) 0.04 mm
- (c) 0.004 mm
- (d) 0.004 mm

Answer: (b)

Solution:

Sol:- Let 1 MSD = 1 mm.

Given 50 VSD = 48 MSD $\therefore 1 \text{ VSD} = \frac{24}{25} \text{ mm}$

$\therefore 1 \text{ VSD} = \frac{48}{50} \text{ MSD}$

$= \frac{24}{25} \text{ MSD}$

Now, L.C = 1 MSD - 1 VSD $\therefore \text{LC} = 0.04 \text{ mm}$

$= 1 - \frac{24}{25} = \frac{1}{25} \text{ mm}$

Question: The 5th harmonic of closed organ pipe coincides with the 1st harmonic of open pipe. Find the ratio of t_c/t_o .

Options:

- (a) $4/3$
- (b) $2/3$
- (c) $5/2$
- (d) $3/4$

Answer: (c)

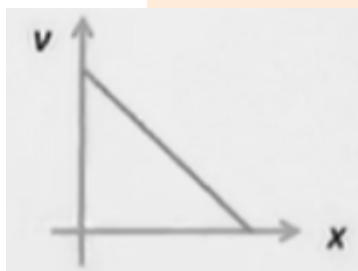
Solution:

$$\text{For closed pipe } f_5 = \frac{5v}{4l_c}$$

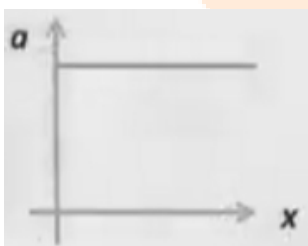
$$\text{For open pipe } f_1 = \frac{v}{2l_o}$$

$$\frac{5v}{4l_c} = \frac{v}{2l_o} \Rightarrow \frac{l_c}{l_o} = \frac{5}{2}$$

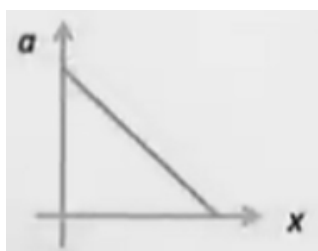
Question: Velocity of particle varies with position as shown in the below graph. Find the correct variation of acceleration with position.



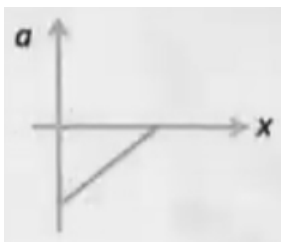
Options:



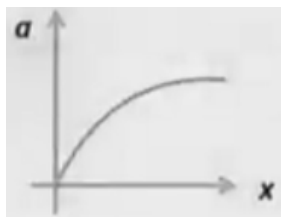
(a)



(b)



(c)

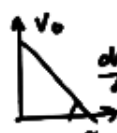


(d)

Answer: (c)

Solution:

Ans:- $a = \frac{dv}{dt} \Rightarrow v \frac{dv}{dx}$



$$\frac{dv}{dx} = \left(-\frac{v_0}{x_0}\right) \Rightarrow \frac{v}{v_0} + \frac{x}{x_0} = 1$$

$$\frac{v}{v_0} = 1 - \frac{x}{x_0}$$

$$v = \frac{v_0}{x_0} (x_0 - x)$$

$$v = -\frac{v_0}{x_0} (x - x_0)$$

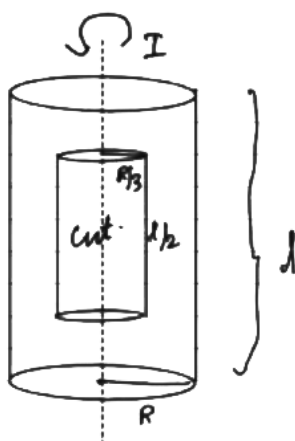
$$a = \frac{dv}{dt} \Rightarrow v \frac{dv}{dx}$$

$$= -\frac{v_0}{x_0} (x - x_0)$$

$$= \left(\frac{v_0}{x_0}\right)^2 (x - x_0)$$

$$a = -ve$$

Question: A solid cylinder of radius $R/3$ and length $l/2$ is cut from a solid cylinder of radius ' R ' and length ' l '. Find the ratio of moment of inertia of new solid to old solid about axis passing through centre & parallel to length.



Options:

- (a) $\frac{161}{162}$
 (b) $\frac{161}{81}$
 (c) $\frac{161}{81}$
 (d) $\frac{82}{81}$

Answer: (a)

Solution:

$$\frac{I_{\text{new}}}{I_{\text{old}}} = ?$$

Initially:

$$\text{Now:- Initially; } I_{\text{old}} = \frac{MR^2}{2}$$

Let mass of cylinder be M .

$$\& \text{ Volume mass density } \Rightarrow \rho = \frac{M}{V}$$

$$\therefore M = \rho \cdot \pi R^2 l$$

Now mass of removed cylinder = $\rho \times \text{Vol. of rem. cylinder.}$

$$= \frac{M}{\pi R^2 l} \times \pi \left(\frac{R}{3}\right)^2 \left(\frac{l}{2}\right)$$

$$= \frac{M}{18}$$

$\therefore$ Moment of inertia of removed cylinder :-

$$I_{\text{removed}} = \frac{1}{2} \times \frac{M}{18} \times \left(\frac{R}{3}\right)^2 = \frac{MR^2}{324}$$

Thus, M.I of remaining i.e.,

$$I_{\text{new}} = I_{\text{old}} - I_{\text{rem}} = \frac{MR^2}{2} - \frac{MR^2}{324}$$

$$I_{\text{new}} = \frac{161MR^2}{324}$$

$$\therefore \frac{I_{\text{new}}}{I_{\text{old}}} = \frac{\frac{161MR^2}{324}}{\frac{MR^2}{2}} = \frac{161}{162}$$

Question: An electron make transition from higher energy orbit (n_2) to lower energy orbit (n_1) in Li^{+2} ion such that $n_1 + n_2 = 4$ & $n_2 - n_1 = 2$. Determine the wavelength emitted in the dimension (in nm)

Options:

- (a) 11.4 nm
- (b) 12.9 nm
- (c) 9.2 nm
- (d) 16.7 nm

Answer: (a)

Solution:

$$\begin{aligned} \text{Ans. } n_1 + n_2 &= 4 \\ n_2 - n_1 &= 2 \\ \Rightarrow n_2 &= 3 \Rightarrow n_1 = 1 \\ \Delta E &= 13.6 Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right) \\ &= 13.6 \times 3^2 \times \left(\frac{1}{1^2} - \frac{1}{3^2} \right) \\ &= 13.6 \times 9 \times \frac{8}{9} \\ &= 13.6 \times 8 \text{ eV} \\ \Delta E &= \frac{hc}{\lambda} \\ \lambda &= \frac{hc}{\Delta E} = \frac{1240 \text{ eV} \cdot \text{nm}}{13.6 \times 8 \text{ eV}} \\ &= 11.4 \text{ nm} \end{aligned}$$

Question: A bob of mass 'm' is connected to a massless inextensible string of length 1 m. It is suspended as shown and released from an angle of 30° with vertical, Find the velocity of bob at Lowest point ($g = 10 \text{ m/s}^2$)

Options:

- (a) 1 m/s
- (b) 1.6 m/s
- (c) 2.6 m/s
- (d) 3 m/s

Answer: (b)

Solution:

Sol:-

Here, work is done only by gravitational force. $\frac{1}{2}mv^2 = mgh$

As it is conservative $\frac{1}{2}mv^2 = mgh$

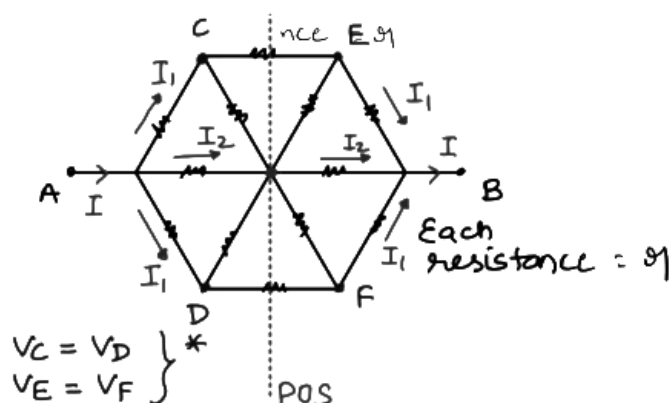
$m \cdot \cancel{E_i} = m \cdot \cancel{E_f}$

$\cancel{K.E_i} + P.E_i = \cancel{K.E_f} + P.E_f$

$m \cdot g \cdot \left(1 - \frac{\sqrt{3}}{2}\right) = \frac{1}{2}mv^2$

$v = \sqrt{20 - 10\sqrt{3}} = \sqrt{2.68} \approx 1.637 \text{ m/s}$

Question: A regular hexagon is formed by six identical resistors, each of resistance R. Find the equivalent resistance between two opposite vertices A and B of the hexagon.

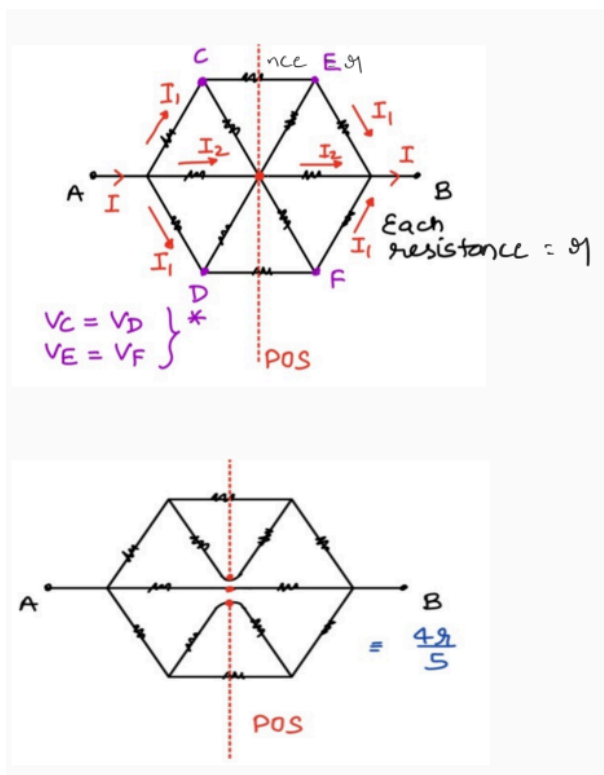


Options:

- (a) $4R/5$
- (b) $R/2$
- (c) $3R/2$
- (d) $2R$

Answer: (a)

Solution:



Question: A metal surface has a work function $\phi = 2.4 \text{ eV}$ is illuminated by two different radiations of energies 4 eV & 6 eV . What is the ratio of wavelength of emitted photons.

Options:

- (a) $2/3$
- (b) $3/2$
- (c) $9/4$
- (d) $4/9$

Answer: (c)

Solution:

Sol: Given work function; $\phi = 2.4 \text{ eV}$

$E_1 = 4 \text{ eV}$; $E_2 = 6 \text{ eV}$.

from photo electric eqn:-

$$\text{Energy of emitted photon} = E - \phi.$$

$$\therefore E_1 = 4 - 2.4 = 1.6 \text{ eV} \quad \left| \quad E_2 = 6 - 2.4 = 3.6 \text{ eV} \right.$$

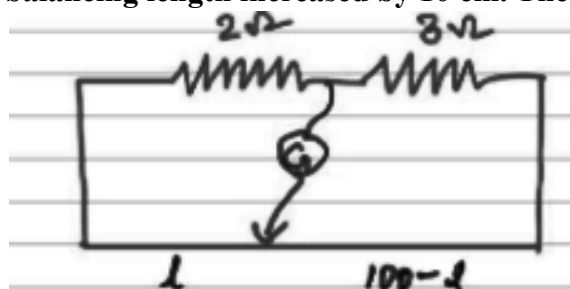
Also we know:-

$$E = \frac{12400}{\lambda} \text{ eV}.$$

$$\therefore \frac{E_1}{E_2} = \frac{\lambda_2}{\lambda_1} = \frac{1.6}{3.6} \Rightarrow \frac{\lambda_1}{\lambda_2} = \frac{36}{16}$$

$$\therefore \frac{\lambda_1}{\lambda_2} = \frac{9}{4}$$

Question: A metre bridge has balancing length 'l' for resistance $2\ \Omega$ & $3\ \Omega$ on left & right segments respectively. When a shunt resistance of 'X' Ω is connected to $3\ \Omega$; balancing length increased by 10 cm. Then find X =



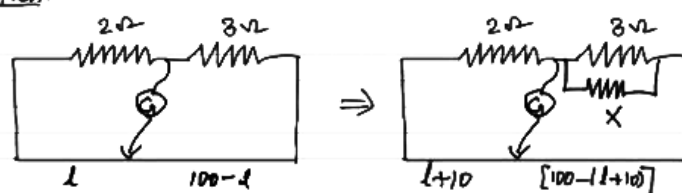
Options:

- (a) $6\ \Omega$
- (b) $4\ \Omega$
- (c) $3\ \Omega$
- (d) $2\ \Omega$

Answer: (a)

Solution:

Sol: Given:-



Using Wheatstone bridge balance condition:-

$$\frac{2}{l} = \frac{3}{100-l}$$

$$5l = 200$$

$$\boxed{l = 40\text{cm}}$$

Now:-

$$\frac{2}{l+10} = \frac{3x}{3+x}$$

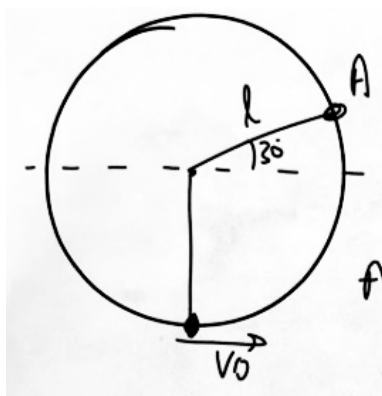
$$\Rightarrow \frac{2}{50} = \frac{3x}{3+x}$$

$$\therefore x = \frac{3x}{3+x}$$

$$6+2x = 3x \Rightarrow \boxed{x = 6\sqrt{2}}$$

Integer type

Question: Find the speed at the lowest point, so that tension in string becomes zero at point A.



Options:

- (a) $\sqrt{\frac{7gl}{2}}$
- (b) $\sqrt{\frac{9gl}{2}}$
- (c) $\sqrt{3gl}$
- (d) $\sqrt{\frac{gl}{2}}$

Answer: (a)

Solution:

Ans:- By conservation of energy between point A and B;

$$\frac{1}{2}mv_0^2 = mg\left(l + \frac{l}{2}\right) + \frac{1}{2}mv_A^2$$

$$v_0^2 = 2g\left(\frac{3l}{2}\right) + v_A^2$$

Now; At the point A: $mg \cos 60^\circ = \frac{mv_A^2}{l}$

$$\Rightarrow v_A^2 = g \times \frac{1}{2} \times 2 \Rightarrow \frac{gl}{2}$$

$$v_0^2 = 3gl + \frac{gl}{2}$$

$$v_0^2 = \frac{7gl}{2}$$

$$v_0 = \sqrt{\frac{7gl}{2}}$$

Question: The current passing through the galvanometer of resistance 100Ω is 5 mA and it becomes 1 mA after a shunt added. Find the resistance of shunt.

Options:

- (a) 25
- (b) 50Ω
- (c) 75Ω
- (d) 100Ω

Answer: (a)

Solution:

sol:- total current $I = 5 \text{ mA} = 5 \times 10^{-3} \text{ A}$

$R_g = 100 \Omega$

current through galvanometer $I_g = 1 \text{ mA} = 1 \times 10^{-3} \text{ A}$

$R_s = ?$

current through the shunt $I_s = I - I_g$

$I_s = 4 \text{ mA} = 4 \times 10^{-3} \text{ A}$

*parallel voltage condition

$I_g \times R_g = I_s \times R_s$

$R_s = \frac{I_g \times R_g}{I_s} = \frac{1 \times 10^{-3} \text{ A} \times 100 \Omega}{4 \times 10^{-3} \text{ A}} = 25 \Omega$

Question: A converging lens has a focal length of +15 cm. An object with a height of 4 cm is placed 20 cm to the left of the lens. Using a cartesian coordinate system where the lens is at origin, calculate the position of the image along the x-axis.

Options:

- (a) 15 cm
- (b) 12 cm
- (c) 20 cm
- (d) 5 cm

Answer: (b)

Solution:

$f = +15 \text{ cm}$

$u = -20 \text{ cm}$

$h_o = 4 \text{ cm}$

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

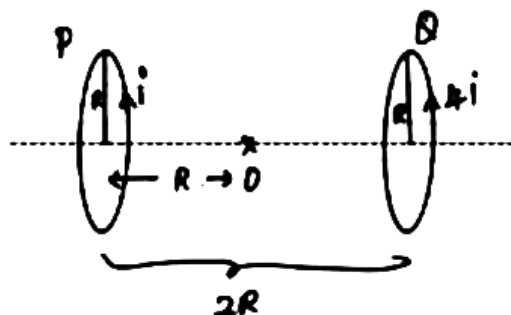
$\frac{1}{v} - \frac{1}{(-20)} = \frac{1}{15}$

$\frac{1}{v} = \frac{1}{15} - \frac{1}{20} = \frac{8-6}{120} = \frac{2}{120}$

$v = 60 \text{ cm}$

$m = \frac{60}{-20} = -3$ $\left| \frac{h_i}{h_o} = -\frac{v}{u} \right| \cdot h_i = 4 \times \left(-\frac{60}{-20} \right) = 4 \times (3) = 12 \text{ cm}$

Question: Two rings of radius are separated by a distance '2R' as shown in fig. Then ring 'P' carries current 'i' & the ring Q carries 4i. Identify the direction of magnetic field due to two rings at mid point of distance between rings.



Options:

- (a) Towards P
- (b) Towards Q
- (c) 0
- (d) Cannot be determined

Answer: (a)

Solution:

Sol:- Magnetic field due to ring carrying current at a point on axis is

$$B_{axis} = \frac{\mu_0 i R^2}{2(x^2 + R^2)^{3/2}}$$

$$B_P = \frac{\mu_0 i R^2}{2(R^2 + R^2)^{3/2}} = \frac{\mu_0 i R^2}{4\sqrt{2}R^3} = \frac{\mu_0 i}{4\sqrt{2}R} \text{ (left)}$$

$$B_Q = \frac{\mu_0 (4i) R^2}{2(R^2 + R^2)^{3/2}} = \frac{4 \cdot \mu_0 i R^2}{4\sqrt{2}R^3} = \frac{\mu_0 i}{\sqrt{2}R} \text{ (left)}$$

$$\therefore B_{Net} = B_P + B_Q$$

Thus Net magnetic field will be towards left.

i.e., Towards P.

Question: A soap bubble of diameter 7 cm and 11 cm diameter is increased to 14 cm. If change in surface energy $(15000 - x)\mu\text{J}$. Find x. ($s = 0.09 \text{ N/m}$)

Options:

- (a) 208
- (b) 216
- (c) 432
- (d) 512

Answer: (b)

Solution:

change in surface energy is given

$$\text{by } \Rightarrow \Delta U = 8\pi(R_2^2 - R_1^2) \times S.$$

$$\Rightarrow 8 \times 3.14 \times 4 \times 10^{-2} (147) \times 10^{-4}$$

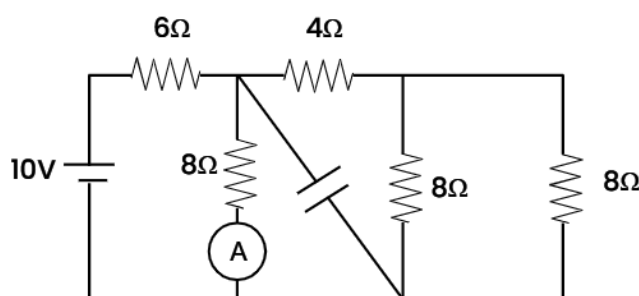
$$\Rightarrow 8 \times 3.14 \times 4 \times 147 \times 10^{-6}$$

$$\Rightarrow 14770.56 \times 10^{-6} \text{ J}$$

$$\Rightarrow (15000 - 229.44) \times 10^{-6} \text{ J}$$

$$\approx \boxed{x = 216} \text{ (closest number)}$$

Question: Find current through ammeter (in A).

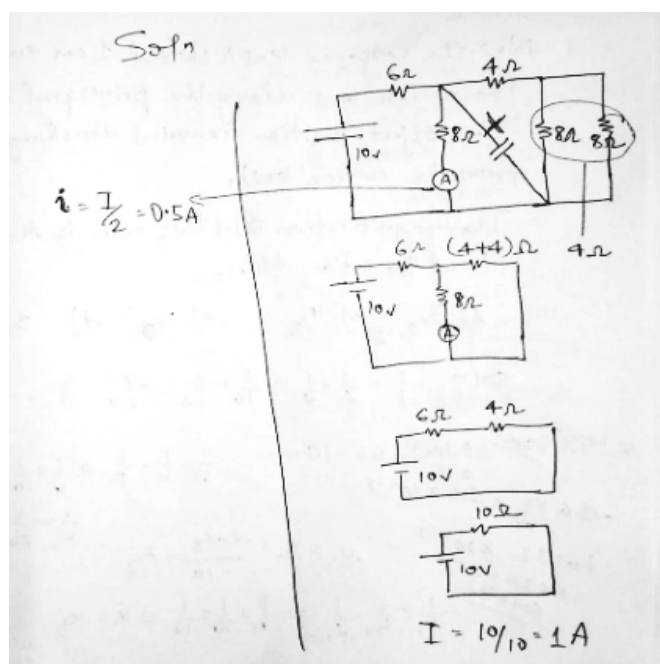


Options:

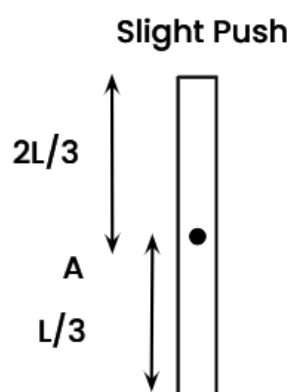
- (a) 1
- (b) 0.5
- (c) 2
- (d) 0.75

Answer: (a)

Solution:



Question: Find when rod becomes horizontal.



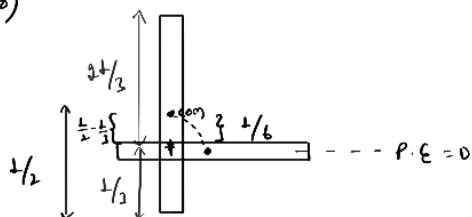
Options:

- (a) $\sqrt{\frac{3g}{L}}$
- (b) $\sqrt{\frac{2g}{L}}$
- (c) $\sqrt{\frac{g}{L}}$
- (d) $\sqrt{\frac{5g}{L}}$

Answer: (a)

Solution:

Q 18)



Sol:- As the work is done here by gravitational force alone.

$$K.E_i + P.E_i = K.E_f + P.E_f$$

$$mg \frac{L}{6} = \frac{1}{2} I \omega^2$$

$$\frac{m \times g \times L}{6} = \frac{1}{2} \times \frac{m L^2}{12} \times \omega^2$$

$$\omega = \sqrt{\frac{3g}{L}}$$

option A.

$$\begin{aligned} I &= I_c + m d^2 \\ I &= \frac{m L^2}{12} + \frac{m L^2}{36} \\ I &= m L^2 \left(\frac{4+1}{36} \right) \\ I &= m L^2 / 9 \end{aligned}$$

Question: A cube of side 8 cm having density 600 kg/m^3 is floating in a liquid of density 900 kg/m^3 . Find height of cube inside liquid.

Options:

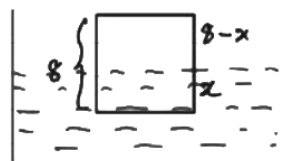
- (a) $\frac{8}{3} \text{ cm}$
 (b) $\frac{16}{3} \text{ cm}$
 (c) $\frac{4}{3} \text{ cm}$
 (d) $\frac{2}{3} \text{ cm}$

Answer: (b)

Solution:

From laws of floatation.

$$\frac{V_{in}}{V} = \frac{\rho_s}{\rho_l}$$



$$\frac{A \times h_{in}}{A \times h} = \frac{600}{900}$$

$$\Rightarrow x = \frac{16}{3} \text{ cm}$$

$$\frac{x}{8} = \frac{6}{9}$$

Question: When monochromatic light of wavelength λ is incident on a photoelectric emitter, the stopping potential is 3.2 V. When light of wavelength 2λ is incident on the same emitter, the stopping potential becomes 0.7 V. Determine the value of λ .

Options:

- (a) 242 nm
- (b) 248 nm
- (c) 348 nm
- (d) 148 nm

Answer: (b)

Solution:

$$\begin{aligned}
 2.5 \text{ eV} &= \frac{hc}{2\lambda} & K E_{\max} &= \frac{hc}{\lambda} - \phi \\
 2.5 \text{ eV} &= \frac{1}{2} \left(\frac{1240}{\lambda(\text{nm})} \right) & 3.2 \text{ eV} &= \frac{hc}{\lambda} - \phi \\
 & & 0.7 \text{ eV} &= \frac{hc}{2\lambda} - \phi
 \end{aligned}$$

$$\lambda = 248 \text{ nm}$$

Question: Two nucleon reactions are given below:



Find stability order of He_2^3 , He_2^4 , He_2^5 .

Options:

- (a) $\text{He}_2^5 > \text{He}_2^3 > \text{He}_2^4$
- (b) $\text{He}_2^3 > \text{He}_2^4 > \text{He}_2^5$
- (c) $\text{He}_2^4 > \text{He}_2^5 > \text{He}_2^3$
- (d) $\text{He}_2^4 = \text{He}_2^5 > \text{He}_2^3$

Answer: (c)

JEE-Main-24-01-2026 (Memory Based)
[EVENING SHIFT]
Chemistry

Question: An electron make transition from higher energy orbit (n_2) to lower energy orbit (n_1) in Li^{2+} ion such that $n_1 + n_2 = 4$ & $n_2 - n_1 = 2$. Determine the wavelength emitted in the dimension (in nm)

Options:

- (a) 12.9 nm
- (b) 11.4 nm
- (c) 16.7 nm
- (d) 9.2 nm

Answer: (b)

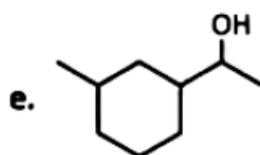
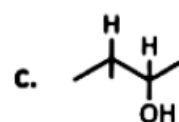
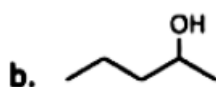
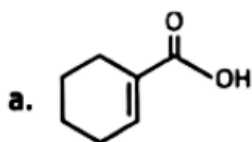
Question: Choose the correct order of second IE of O, C, N and F

Options:

- (a) $\text{C} < \text{N} < \text{F} < \text{O}$
- (b) $\text{C} < \text{F} < \text{O} < \text{N}$
- (c) $\text{C} < \text{N} < \text{O} < \text{F}$
- (d) $\text{C} < \text{O} < \text{F} < \text{N}$

Answer: (a)

Question: Which of the following molecules is secondary alcohol?



Options:

- (a) b, c, e only
- (b) b, c, d, e only
- (c) a, c, d, e only
- (d) a, b, d only

Answer: (a)

Question: How many linear tripeptides are possible with valine (Val), Glycine (Gly) and Alanine (Ala). No amino acid should be repeated?

Options:

- (a) 8
- (b) 5
- (c) 6
- (d) 4

Answer: (c)

Question: Order of wavelength of absorbed radiation for the below given complexes is,

- (a) $[\text{Co}(\text{NH}_3)_6]^{3+}$
- (b) $[\text{Co}(\text{NH}_3)_5\text{H}_2\text{O}]^{3+}$
- (c) $[\text{CoF}_6]^{3-}$
- (d) $[\text{Co}(\text{CN})_6]^{3-}$

Options:

- (a) $d > a > c > b$
- (b) $d > a > b > c$
- (c) $d < a < b < c$
- (d) $d < a < c < b$

Answer: (b)

Question: Given:

$$\Delta H_{\text{atom}}(\text{CH}_4) = x \text{ kJ mole}^{-1}$$

$$\Delta H_{\text{atom}}(\text{C}_2\text{H}_6) = y \text{ kJ mole}^{-1}$$

Find out bond energy (C - C) (kJ/mole)

Options:

- (a) $y - x$
- (b) $y - 4x$
- (c) $y - \frac{3x}{2}$
- (d) $y - 2x$

Answer: (c)

Question: Which of the following have same bond order and are paramagnetic?

Options:

- (a) $\text{O}_2^+, \text{N}_2^-$
- (b) $\text{O}_2^+, \text{O}_2^-$
- (c) $\text{O}_2^-, \text{N}_2^-$
- (d) $\text{O}_2^-, \text{N}_2^+$

Answer: (c)

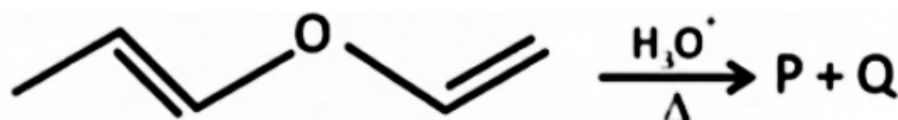
Question: An electron make transition from higher energy orbit (n_2) to lower energy orbit (n_1) in Li^{2+} ion such that $n_1 + n_2 = 4$ & $n_2 - n_1 = 2$. Determine the wavelength emitted in the dimension (in nm)

Options:

- (a) 12.9 nm
- (b) 11.4 nm
- (c) 16.7 nm
- (d) 9.2 nm

Answer: (b)

Question: How would you distinguish product formed P and Q of reaction given below



Options:

- (a) Fehling solution test
- (b) Tollens test
- (c) 2, 4 DNP test
- (d) Iodoform test

Answer: (d)

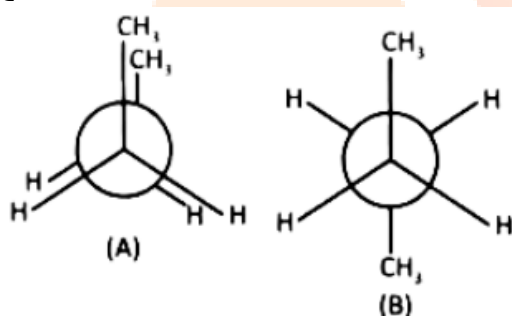
Question: A group VII element which has a + 7 oxidation state forms a salt with potassium (K). What is the colour of this salt?

Options:

- (a) Green
- (b) Yellow
- (c) Orange
- (d) Purple

Answer: (d)

Question: Given below are two statements based on structures given



Statement-I: B is more stable than A.

Statement-II: Dihedral angle of B is more than A.

In the light of the above two statements, choose the correct option.

Options:

- (a) Both Statement-I and Statement-II are correct
- (b) Both statement-I and statement-II are incorrect
- (c) Statement-I is correct but statement-II is incorrect
- (d) Statement-I is incorrect but statement-II is correct

Answer: (a)

Question: Vapor pressure of two volatile species A and B are 55 mm and 120 mm respectively. If mole fraction of 'A' in liquid state is 0.8, then mole fraction of 'B' in vapor state is

Options:

- (a) 0.65
- (b) 0.45
- (c) 0.35
- (d) 0.53

Answer: (c)

Question: Find incorrect statement among the following

Options:

- (a) C^{13} is radioactive isotope
- (b) Covalency of carbon greater than 4 is possible
- (c) Carbon can exhibit +2 & +4 oxidation state
- (d) In group-14, CO_2 is most acidic

Answer: (b)

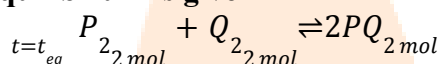
Question: Which of the following statement is correct about resonance and resonating structures?

Options:

- (a) Resonating structure with more covalent bonds is more stable
- (b) The resonance structure differ in position of electrons and relative position of atoms
- (c) The stability of resonance hybrid decreases with increasing number of equivalent resonating structure
- (d) Electronegative atom bearing positive charge in the canonical form is more stable

Answer: (a)

Question: Consider the following reversible reaction wherein the moles of species at equilibrium is given

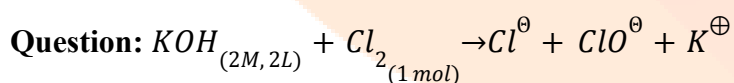


If one mole of P_2 and one mole of Q_2 are added at equilibrium. The number of moles of P_2 , Q_2 and PQ at new equilibrium, respectively are

Options:

- (a) $\frac{4}{3}, \frac{4}{3}, \frac{8}{3}$
- (b) $\frac{8}{3}, \frac{8}{3}, \frac{8}{3}$
- (c) $\frac{4}{3}, \frac{4}{3}, \frac{4}{3}$
- (d) $\frac{8}{3}, \frac{8}{3}, \frac{4}{3}$

Answer: (b)



Find the conc. Of each product and choose correct option

Options:

- (a) $[Cl^-] = [ClO^-] = [K^+] = 0.5\text{ M}$
- (b) $[Cl^-] = [K^+] = 1.5\text{ M}$
- (c) $[Cl^-] = [ClO^-] = 0.5\text{ M}$
- (d) $[Cl^-] = [ClO^-] = 0.75\text{ M}$

Answer: (c)

Question: Given below are two statements.

Statement I: Two different aldehydes on cross aldol condensation always give four products.

Statement II: Among benzaldehyde and acetophenone, only acetophenone reacts with semicarbazide.

In the light of the above two statements, choose the correct option.

Options:

- (a) Both statement I and statement II are correct
- (b) Both statement I and statement II are incorrect
- (c) Statement I is correct but statement II is incorrect
- (d) Statement I is incorrect but statement II is correct

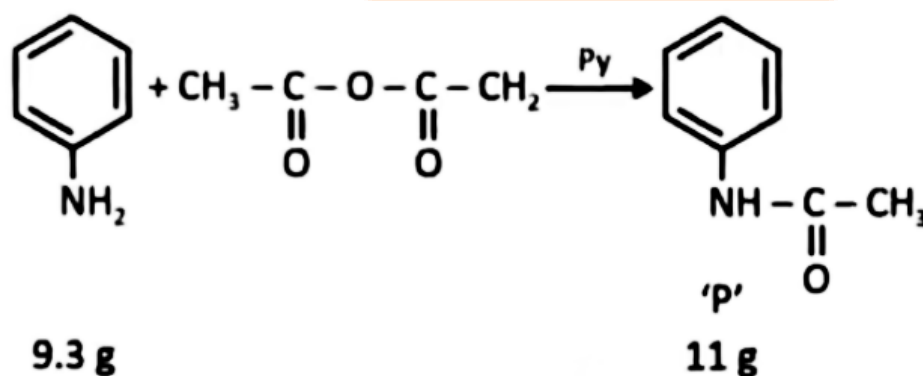
Answer: (b)

Question: In a first order reaction, $t_{1/2} = 245$ days of compound 'A'. After x days 75% of 'A' remains, then calculate the value of 'x'. (Take $\log 2 = 0.3$ and $\log 3 = 0.48$)

Options:

Answer: (98)

Question: Consider the following is (nearest integer)



If % yield of reaction is x, the value of x/10 is (nearest integer)

Options:

Answer: (81)

Question: A compound $\text{Cr}(\text{H}_2\text{O})_6 \cdot \text{Cl}_3$ show conductance similar to 1 : 2 electrolyte in aq.

Solution. 9.6 g of this complex is passed through a cation exchanger then excess of AgNO_3 solution is added. Find mass of AgCl precipitated in gram?

[Molar of Cr = 52, Cl = 35.5]

Options:

Answer: (10)

Question: 0.18 M HQ solution has molar conductivity $\frac{1}{30}$ times the molar conductivity of 0.02 M HZ solution.

Find the value of $\text{p}K_a(\text{HQ}) - \text{p}K_a(\text{HZ})$. [Given that $\alpha \ll 1$]

Assume that $\lambda_m^\infty Q^\ominus = \lambda_m^\infty Z^\ominus$

Options:

Answer: (2)

JEE-Main-24-01-2026 (Memory Based)

[EVENING SHIFT]

Maths

Question: If words arranged in a dictionary alphabetically then find the rank of UDAIPUR.

Answer: (1922)

6	2	1	2	4	6	5
U	D	A	I	P	U	R
5	1	0	0	0	1	0
$\frac{6!}{2!}$	5!	4!	3!	2!	1!	0!

$$\Rightarrow \frac{5(6!)}{2} + 1(5!) + 0(4!) + 0(3!) + 0(2!) + 1(1!) + 0(0)$$

$$\Rightarrow \frac{5(720)}{2} + 120 + 0 + 0 + 0 + 1$$

$$\Rightarrow \frac{3600}{2} + 120$$

$$1800 + 120 + 1 = 1921 + 1 = 1922$$

Question: Maximum value of n for which 40^n circulates $60!$ = ?

Answer: (18)

$$40 = 2^3 \times 5$$

$$40^n = 2^{8n} 5^n$$

$$\sum n(n!) = \sum_{k=1}^{\infty} \left[\frac{n}{p^k} \right]$$

$$\text{Exporment 1:-} \left[\frac{60}{5} \right] + \left[\frac{60}{25} \right]$$

$$= 12 + 2 = 14$$

$$\text{Exporment 2:-} \left[\frac{60}{2} \right] + \left[\frac{60}{4} \right] + \left[\frac{60}{8} \right] + \left[\frac{60}{16} \right] + \left[\frac{60}{32} \right]$$

$$= 30 + 15 + 7 + 3 + 1$$

$$= 56$$

Request power 56

$$3n = 56$$

$$n = \frac{56}{3} \simeq 18.86$$

$$n = 18$$

Question: The image of parabola $x^2 = 4y$ in the line $x - y = 1$ is

Options:

(a) $(y - 1)^2 = 4(x + 1)$

(b) $(y + 1)^2 = 4(x - 1)$

(c) $(y + 1)^2 = 4(x + 1)$

(d) $(y - 1)^2 = 4(x - 1)$

Answer: (b)

Let $P(-2t, t^2)$

$$\frac{x-2t}{1} = \frac{y-t^2}{-1} = \frac{2(2t-t^2-1)}{2}$$

$$x = -2t + t^2 + 1 + 2t : y - t^2 = 2t - t^2 - 1$$

$$x = t^2 + 1 \quad y = 2t - 1$$

$$x - 1 = \left(\frac{y+1}{2}\right)^2$$

$$4(x - 1) = (y + 1)^2$$

Question: Let the equation $x^4 - ax^2 + 9 = 0$ have four real and distinct roots then the minimum integral value of a ?

Answer: (6)

$$ax^2 = x^4 + 9$$

$$a = x^2 + \frac{9}{x^2}$$

$$Am \geq Gm$$

$$\Leftrightarrow \frac{x^2 + \frac{9}{x^2}}{2} \geq \sqrt{x^2 \cdot \frac{9}{x^2}}$$

$$x^2 + \frac{9}{x^2} \geq 6$$

$$\therefore a \geq 6$$

Minimum integral value of a is 6

Question: $Z = (1 + i)(1 + 2i)(1 + 3i) \dots (1 + ni)$; $n \in \mathbb{N}$ and $|z|^2 = 44200$ then n is = ?

Answer: (n=5)

$$|z| = |1 + i||1 + 2i||1 + 3i| \dots |1 + ni|$$

$$|z|^2 = (2)(5)(10) \dots (1 + n^2)$$

$$\text{Given } |z|^2 = 44200$$

$$\text{Check } n = 1(1 + 1^2) = 2$$

$$n = 2(1 + 2^2) = 5$$

$$n = 3(1 + 9) = 10$$

$$n = 4(1 + 16) = 17$$

$$n = 5(1 + 25) = 26$$

$$2 \times 5 \times 10 \times 17 \times 26 = 44200,$$

$$\text{so } n = 5$$

$$s = \left(\frac{1}{3} + \frac{4}{7}\right) + \left(\left(\frac{1}{3}\right)^2 + \left(\frac{4}{7}\right)^2 + \left(\frac{1}{3}\right)\left(\frac{4}{7}\right)\right)$$

Question: Value of sum $+\left(\left(\frac{1}{3}\right)^3 + \left(\frac{1}{3}\right)^2\left(\frac{4}{7}\right) + \left(\frac{1}{3}\right)\left(\frac{4}{7}\right)^2 + \left(\frac{4}{7}\right)^3\right) + \dots$

Options:

(a) $\frac{3}{2}$

(b) $\frac{5}{2}$

(c) $\frac{1}{2}$

(d) 2

Answer: (b)

$a = \frac{1}{3}, b = \frac{4}{7}$

$s(a+b) + (a^2 + ab + b^2) + (a^3 + a^2b + ab^2 + b^3) - \dots$

$T_n = \frac{a^{k+1} - b^{k+1}}{a-b}$

$s = \sum_{k=1}^{\infty} \frac{a^{k+1} - b^{k+1}}{a-b}$

$\frac{1}{a-b} \left[\sum_{k=1}^{\infty} a^{k+1} - \sum_{k=1}^{\infty} b^{k+1} \right]$

$\frac{1}{a-b} [(a^2 + a^3 + \dots) - (b^2 + b^3 + \dots)]$

$\frac{1}{a-b} \left[\frac{a^2}{1-a} - \frac{b^2}{1-b} \right]$

$\frac{1}{\frac{1}{3} - \frac{4}{7}} \left[\frac{\left(\frac{1}{3}\right)^2}{\frac{2}{3}} - \frac{\left(\frac{4}{7}\right)^2}{1 - \frac{4}{7}} \right]$

$= \frac{21}{5} \left[\frac{\frac{1}{9}}{\frac{2}{3}} - \frac{\frac{16}{49}}{\frac{3}{7}} \right] - \frac{21}{5} \left[\frac{7}{42} - \frac{32}{42} \right] \Rightarrow -\frac{21}{5} \times \frac{-25}{42} = \frac{5}{2} \Rightarrow s = \frac{5}{2}$

Question: Domain of $\sin^{-1}\left(\frac{1}{x^2 - 2x - 1}\right)$ is $(-\infty, \alpha] \cup [\beta, \delta] \cup [\lambda, \infty)$ the value of $\alpha + \beta + \delta + \lambda = ?$

Options:

(a) 17

(b) 4

(c) 3

(d) 6

Answer: (b)

$-1 \leq \frac{1}{x^2 - 2x - 1} \leq 1$

$x^2 - 2x - 1 \in (-\infty, -1] \cup [1, \infty)$

$x^2 - 2x - 1 \leq -1 \& x^2 - 2x - 1 \geq 1$

$x^2 - 2x \leq 0 \quad x^2 - 2x - 2 \geq 0$

$x \in [0, 2] \quad x \in (-\infty, 1 - \sqrt{3}] \cup [1 + \sqrt{3}, \infty)$

$x \in (-\infty, 1 - \sqrt{3}] \cup [0, 2] \cup \{1 + \sqrt{3}, \infty)$

$1 \cdot \sqrt{3} + 0 + 2 + 1 + \sqrt{3} = 4$

Question: $f(x) = |\ln x| - |x - 1|$

(i) $f(x)$ is different from all $n > 0$

(ii) $f(x)$ is increasing in $(1, \infty)$

(iii) $f(x)$ is decreasing in $(0, 1)$

Options:

(a) (i) and (ii) is there

- (b) (i), (ii) and (iii) are correct
 (c) (i) and (ii) are correct
 (d) (i), (ii) and (iii) are incorrect

Answer: (a)

$$f(x) = \begin{cases} \ln x - x + 1; & x \geq 1 \\ -\ln x + n - 1; & 0 < x < 1 \end{cases}$$

$$f'(x) = \begin{cases} \frac{1}{x} - 1, & x \geq 1 \\ -\frac{1}{x} + 1, & 0 < x < 1 \end{cases}$$

$$f'(1^+) = f'(1^-) = 0$$

$$f''(x) < 0 \quad \forall n > 1 \quad 0 < x < 1$$

$$\frac{x}{\alpha} = \frac{y+1}{2} = \frac{z-4}{-\alpha}$$

$$\frac{x-1}{\alpha} = \frac{y-2}{8} = \frac{z-1}{-2\alpha}$$

Question: Shortest distance is $\sqrt{2}$ then $\alpha = ?$

Answer: (0)

$$\frac{x}{\alpha} = \frac{y+1}{2} = \frac{z-4}{-\alpha}$$

$$A(0, -1.4), \bar{b}_1(\alpha, 2, -\alpha)$$

$$\frac{x-1}{\alpha} = \frac{y-2}{8} = \frac{z-1}{-2\alpha}$$

$$\bar{B}(1, 2, 1)\bar{b}_2(\alpha, 8, -2\alpha)$$

$$\overline{AB} = \hat{i} + 3\hat{j} - 3\hat{k}$$

$$\bar{b}_1 \times \bar{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \alpha & 2 & -\alpha \\ \alpha & 8 & -2\alpha \end{vmatrix}$$

$$\alpha = \frac{8 \pm \sqrt{7056 - 2576}}{19}$$

$$\frac{84 \pm \sqrt{4480}}{14}$$

$$\bar{b}_1 \times \bar{b}_2 = 4\alpha\hat{i} + \alpha\hat{j} + 6\alpha\hat{k}$$

$$\alpha = \frac{84 \pm 8\sqrt{70}}{14}$$

$$|\bar{b}_1 \times \bar{b}_2| = \sqrt{16\alpha^2 + \alpha^4 + 36\alpha^2}$$

$$= \sqrt{\alpha^4 + 52\alpha^2}$$

$$\alpha = \frac{42 \pm 4\sqrt{70}}{7}$$

$$|\bar{b}_1 \times \bar{b}_2| = |\alpha|\sqrt{\alpha^2 + 52}$$

$$d = \left| \frac{AB \cdot (\bar{b}_1 \times \bar{b}_2)}{|\bar{b}_1 \times \bar{b}_2|} \right|$$

$$\alpha = 6 \pm \frac{4\sqrt{70}}{7}$$

$$\sqrt{2} = \left| \frac{4\alpha + 3\alpha^2 - 18\alpha}{|\alpha|\sqrt{\alpha^2 + 52}} \right|$$

$$2 = \frac{(3\alpha - 14)}{\alpha^2 + 52}$$

$$2\alpha^2 + 104 = 9\alpha^2 - 84\alpha + 196$$

$$7\alpha^2 - 84\alpha + 92 = 0$$

Question: $\lim_{x \rightarrow 0} \frac{\tan(\tan x) - \tan(\sin x)}{\tan x - \sin x}$

Options:

- (a) 1
 (b) 2
 (c) $\frac{1}{2}$

(d) -1

Answer: (a)

$$\lim_{x \rightarrow 0} \frac{\tan(\tan x) - \tan(\sin x)}{\tan x - \sin x}$$

$$\frac{(\tan(\tan x) + \tan \sin x)}{(\tan(\tan x) + \tan(\sin x))}$$

$$= (1)(1) = 1$$

Question: If $4x^2 + y^2 < 52$, $x, y \in \mathbb{I}$

No of ordered pairs (x, y) is

Options:

(a) 38

(b) 77

(c) 67

(d) 87

Answer: (b)

$$x = 0 \Rightarrow y^2 < 52 \left. \begin{array}{l} -7 \leq y \leq 7 \end{array} \right\} 15 \times 1 = 15$$

$$x = \pm 1 \Rightarrow y^2 < 48 \left. \begin{array}{l} -6 \leq y \leq 6 \end{array} \right\} 13 \times 2 = 26$$

$$x = \pm 2 \Rightarrow y^2 < 36 \left. \begin{array}{l} -5 \leq y \leq 5 \end{array} \right\} 11 \times 2 = 22$$

$$x = \pm 3 \Rightarrow y^2 < 16 \left. \begin{array}{l} -3 \leq y \leq 3 \end{array} \right\} 7 \times 2 = 14$$

$$\text{Total: } 15 + 26 + 22 + 14$$

$$= 77$$

Question: a_1, a_2, a_3, a_4 , AP $a_1 + a_2 + a_3 + a_4 = 48$ $\ell \rightarrow$ c.d what will be largest value in A.P

Answer: (27)

Let $a - 3d, a - d, a + d, a + 3d$

$$a_1 = a - 3d, a_4 = a + 3d (\text{last term})$$

$$l = 2d \Rightarrow \boxed{d = \frac{\ell}{2}}$$

$$S = (a - 3d) + (a - d) + (a + d) + (a + 3d) = 98$$

$$4a = 98$$

$$\boxed{a = 24.5}$$

$$24.5 - 3d, 24.5 - d, 24.5 + d, 24.5 + 3d$$

$$a_1 \cdot a_2 \cdot a_3 \cdot a_4 = \sqrt{361}$$

$$(24.5 - 3d)(24.5 - d)(24.5 + d)(24.5 + 3d) = 361$$

$$(24.5)^2 - 144d^2 - 1296d^2 + 9d^4 + 16d^4 = 361$$

$$2400.25 - 1440d^2 + 25d^4 = 361$$

$$(24.5 - 5d)^2 = 361$$

$$(24.5 - 5d) = \sqrt{361}$$

$$24.5 - 5d = \pm 19$$

$$125 = 5d^2$$

$$d = 5$$

Common difference $\ell = 2d = 10$

$$a - 3d = 24.5 - 15 = 9.5$$

$$a - d = 24.5 - 5 = 19.5$$

$$a + d = 24.5 + 5 = 29.5$$

$$a + 3d = 24.5 + 15 = 39.5$$

27.

Question: If $\int \frac{7x^{10} + 9x^8}{(1 + x^2 + 2x^9)^2} dx = f(x) + C$ and $f(1) = \frac{1}{4}$ then $f(x)$?

Options:

(a) $\frac{x^9}{2x^2 + 9 + x^9}$

(b) $\frac{x^9}{2 + x^2 + x^9}$

(c) $\frac{1 + x^2 + 2x^9}{x^9}$

(d) $\frac{1 + x^9 + 2x^2}{x^9}$

Answer: (c)

$$\int \frac{\left(\frac{9}{x^8} + \frac{9}{x^{10}}\right) dx}{\left(\frac{1}{x^9} + \frac{1}{x^7} + 2\right)^2}$$

$$\text{Let } x^{-9} + x^{-7} + 2 = t$$

$$\Rightarrow -\int \frac{dt}{t^2} = -\int t^{-2} dt = \frac{-t^{-1}}{-1} + C = \frac{1}{t} + C$$

$$\frac{1}{x^{-9} + x^{-7} + 2} + C$$

$$\frac{x^9}{1 + x^2 + 2x^9} + C$$

Question: $x : \{1, 2, 3, \dots, 19\}$

$Y : \{y_i : y_i = ax_i + b, x_i \in x\}$

Mean and variance of y is 30 and 120 respectively then sum of value(s) of b ?

Answer: (50)

$$\frac{\sum y_i}{19} = a \frac{\sum x_i}{19} + b$$

$$\Leftrightarrow 30 = a \cdot \frac{19 \times 20}{19 \times 2} + b$$

$$10a + b = 30$$

Original variance

$$\frac{1^2 + 2^2 + \dots + 19^2}{19} - \left(\frac{\sum x_i}{19}\right)^2$$

$$\frac{19 \times 20 \times 39}{19 \times 6} - 100 = 30$$

$$\Leftrightarrow \text{Variance}(y_i) = 30a^2 = 120$$

$$\Leftrightarrow a^2 = 4$$

$$a = 2 \Rightarrow b = 10$$

$$a = -2 \Rightarrow b = 50$$

Question:

Options:

(a)

(b)

(c)

(d)

Answer: ()

Question:

Options:

(a)

(b)

(c)

(d)

Answer: ()

Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

Question:

Options:

- (a)
- (b)
- (c)



(d)

Answer: 0



Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

Question:

Options:

- (a)
- (b)
- (c)
- (d)

Answer: ()

